



Summary of this week

- This week, you learned:
 - How to derive the six steps of generic Gaussian sequential probabilistic inference.
 - How to apply those six steps to a linear model to derive the standard linear Kalman filter, and the assumptions that are made when doing so.
 - How to interpret the variables and steps of the Kalman filter.
- It might be a good idea at this point to review Week 4 (“State-estimation application of a Kalman filter”) of Course 1 (“Kalman Filter Boot Camp”) to remember:
 - How to implement the linear Kalman filter in Octave code, and
 - Scenarios when the standard KF worked well and how it could fail if the assumptions made during the derivation are violated.



Where to from here?

- Next week, you will learn some techniques to make the Kalman filter “bulletproof” (how to make it work under a wide range of circumstances, including those not anticipated or permitted in its derivation):
 - You will learn how to improve its numeric robustness.
 - You will learn how to increase its precision.
 - You will learn how to initialize and tune a KF.
 - You will learn how to estimate noise biases if they are not zero mean.
 - You will learn how to adjust the filter equations when the process and sensor noises are cross-correlated.
 - You will learn how to adjust the filter equations when the process and/or sensor noises are autocorrelated.



Credits

Credits for photos in this lesson

- “Map” on slide 2: Pixabay license
 (<https://pixabay.com/en/service/license/>), cropped from
<https://pixabay.com/en/compass-navigation-map-direction-390054/>